

Java 13 — Text Blocks (Preview)

Java Version: 13

Year: 2019

Feature: Text Blocks (*Preview*)

UniqueClassName: TextBlocks

Definition

Text Blocks were introduced as a **preview feature in Java 13** to make it easier to write **multi-line string literals**.

Before Java 13, multi-line text generally required:

- `\n` escape sequences
- String concatenation using `+`
- Multiple string literals

For example:

```
String json = "{\n" +  
              "  \"name\": \"Anurag\", \n" +  
              "  \"language\": \"Java\"\n" +  
              "}";
```

With a text block:

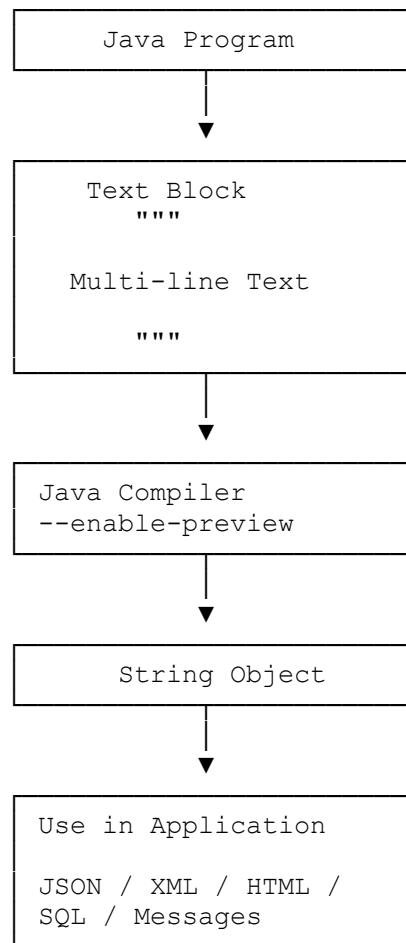
```
String json = ""  
    {  
        "name": "Anurag",  
        "language": "Java"  
    }  
    "";
```

Text blocks are especially useful for:

- JSON
- XML
- HTML
- SQL
- Multiline messages
- Configuration data
- Embedded code snippets

Important: Text blocks were a **preview feature in Java 13**. They were refined in Java 14 and became a permanent Java language feature in Java 15.

Flowchart



Java Code Example

TextBlocks.java

```
public class TextBlocks {  
    public static void main(String[] args) {  
        String message = """  
            Hello,  
            Welcome to Java 13!  
            This is a text block.  
            """;  
        System.out.println(message);  
    }  
}
```

Output

Hello,
Welcome to Java 13!
This is a text block.

Text Block with JSON

One of the most useful applications is writing JSON directly in Java code.

```
public class TextBlocks {  
    public static void main(String[] args) {  
        String json = ""  
        {  
            "id": 101,  
            "name": "Anurag",  
            "language": "Java"  
        }  
        "";  
  
        System.out.println(json);  
    }  
}
```

Output:

```
{  
  "id": 101,  
  "name": "Anurag",  
  "language": "Java"  
}
```

This is significantly more readable than manually adding \n and escaped quotation marks.

Text Block with SQL

```
public class TextBlocks {  
    public static void main(String[] args) {  
        String sql = ""  
        SELECT id, name, email  
        FROM customer  
        WHERE status = 'ACTIVE'  
        ORDER BY name;  
    }  
}
```

```
        """;

        System.out.println(sql);
    }
}
```

Output:

```
SELECT id, name, email
FROM customer
WHERE status = 'ACTIVE'
ORDER BY name;
```

Text Block with HTML

```
public class TextBlocks {

    public static void main(String[] args) {

        String html = ""
            <html>
                <body>
                    <h1>Java 13</h1>
                    <p>Text Blocks</p>
                </body>
            </html>
            """;

        System.out.println(html);
    }
}
```

Text blocks make embedded HTML much easier to read.

Important Syntax

A text block begins and ends with **three double quotes**:

```
String text = ""
    Line 1
    Line 2
    Line 3
    """;
```

Conceptually:

```
""" ← Opening delimiter
|
├── Line 1
├── Line 2
├── Line 3
|
""" ← Closing delimiter
```

Text Block vs Traditional String

Traditional String

```
String json = "{\n" +
    "    \"name\": \"Java\", \n" +
    "    \"version\": 13\n" +
    "}";
```

Text Block

```
String json = """
    {
        "name": "Java",
        "version": 13
    }
    """;
```

The text block is much easier to maintain because the source code visually resembles the resulting text.

Escape Characters

Text blocks still support Java escape sequences.

```
public class TextBlocks {

    public static void main(String[] args) {

        String text = """
            Java
            "Text Blocks"
            Tab:\tValue
            """;

        System.out.println(text);
    }
}
```

For example, `\t` represents a tab.

Compilation in Java 13

Because text blocks were a preview feature in Java 13, compile using:

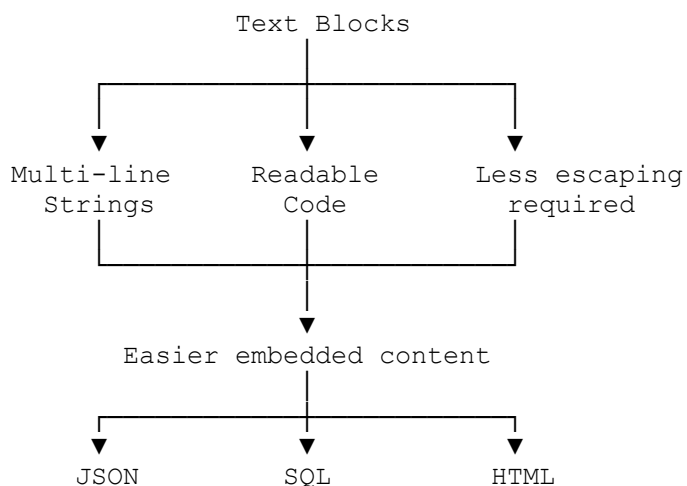
```
javac --enable-preview --release 13 TextBlocks.java
```

Run using:

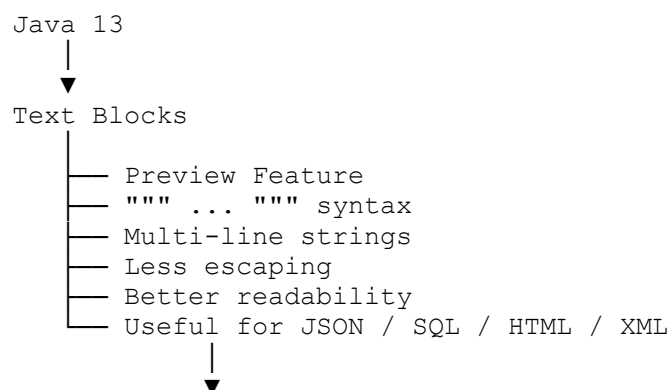
```
java --enable-preview TextBlocks
```

The `--enable-preview` option enables preview language features in Java 13.

Key Benefits



Key Point



Permanent in Java 15

In short: Java 13 introduced **Text Blocks as a preview feature**, providing a cleaner way to represent multi-line strings directly in source code. The feature was refined in Java 14 and finalized in **Java 15**.